

BARRANQUILLA, ColombiaWMO: **800280**Lat: **10.890N**Lon: **74.781W**Elev: **30**StdP: **100.97**Time Zone: **-5.00 (W05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	23.1	19.9	14.6	26.4	20.5	15.2	26.5	13.0	27.0	11.5	26.8	1.9	180	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.2	34.5	27.4	33.9	27.3	33.1	27.1	28.7	32.0	28.3	31.5	27.9	31.1	3.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	30.2	27.5	23.5	29.8	27.1	23.0	29.6	93.4	32.4	91.5	31.7	89.9	31.0	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.4	8.4	DB	20.7	36.8	1.3	1.0	19.8	37.5	19.0	38.1	18.3	38.6	17.3	39.4	(4)
(5)			WB	19.5	29.7	1.3	0.8	18.6	30.2	17.9	30.7	17.2	31.1	16.2	31.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	27.7	27.8	28.3	29.0	29.1	28.9	28.8	28.9	28.7	28.2	28.2	28.2	(6)	
	DBStd	1.19	0.83	0.96	1.02	1.01	1.18	1.11	1.12	1.20	1.34	1.28	1.15	0.92	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6748	549	498	566	569	591	568	584	587	561	564	545	566	(10)	
	CDD18.3	3706	291	265	308	319	333	318	326	328	311	306	295	307	(11)	
	CDH23.3	38805	2787	2527	3043	3381	3652	3537	3603	3641	3330	3108	3020	3178	(12)	
	CDH26.7	13702	803	752	978	1206	1383	1356	1361	1424	1297	1117	1018	1005	(13)	
(14) Wind	WSAvg	3.8	5.1	5.5	5.5	4.7	3.2	2.9	3.3	3.1	2.6	2.4	2.8	4.1	(14)	
(15) Precipitation	PrecAvg	854	0	1	1	25	95	102	78	105	153	167	95	24	(15)	
	PrecMax	1619	7	8	13	142	288	358	359	401	300	363	386	167	(16)	
	PrecMin	411	0	0	0	0	8	1	0	4	23	19	5	0	(17)	
	PrecStd	290	1	2	2	34	66	74	76	74	68	81	76	36	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.2	34.1	34.9	35.0	34.8	34.8	35.0	35.2	34.2	33.6	33.2	(19)	
		MCWB	25.9	25.9	26.7	27.2	27.5	27.5	27.2	27.4	28.0	28.2	27.4	26.9	(20)	
	2%	DB	31.9	32.1	32.9	33.4	33.8	33.4	33.2	33.8	33.9	33.1	32.7	32.2	(21)	
		MCWB	25.5	25.5	26.2	26.7	27.5	27.4	26.9	27.3	27.6	27.6	27.1	26.3	(22)	
	5%	DB	31.1	31.2	31.9	32.8	32.9	32.8	32.8	32.9	32.8	32.1	31.9	31.4	(23)	
		MCWB	25.4	25.3	25.9	26.4	27.2	27.3	26.8	27.1	27.4	27.2	26.8	26.1	(24)	
	10%	DB	30.2	30.3	31.0	31.8	31.9	31.9	31.9	32.0	31.9	31.1	31.0	30.9	(25)	
		MCWB	25.1	25.0	25.5	26.1	26.9	26.9	26.6	26.9	27.0	27.0	26.8	26.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.2	28.2	28.2	29.1	28.7	28.2	28.8	29.1	29.2	28.7	28.5	(27)	
		MCDB	30.0	30.4	31.2	32.2	32.8	32.5	32.0	32.0	32.7	32.3	31.1	31.0	(28)	
	2%	WB	26.6	26.5	27.2	27.5	28.2	28.1	27.7	28.1	28.4	28.4	27.9	27.5	(29)	
		MCDB	29.7	30.0	30.6	31.2	31.6	31.7	31.4	31.5	31.6	31.3	30.6	30.2	(30)	
	5%	WB	26.1	26.0	26.5	27.0	27.6	27.6	27.2	27.6	27.9	27.9	27.4	26.9	(31)	
		MCDB	29.3	29.6	30.1	30.5	30.9	31.1	30.9	30.9	30.9	30.5	30.1	29.8	(32)	
	10%	WB	25.7	25.6	26.0	26.5	27.2	27.2	26.8	27.2	27.5	27.4	26.9	26.4	(33)	
		MCDB	28.8	29.1	29.7	30.0	30.3	30.5	30.2	30.4	30.3	30.0	29.7	29.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.7	6.9	7.2	7.2	7.2	7.2	7.6	7.8	7.4	6.9	6.5	(35)	
		MCDBR	7.3	7.5	7.7	8.1	7.9	8.0	7.9	8.2	8.6	8.1	7.5	7.0	(36)	
	5% WB	MCWBR	3.0	3.0	2.9	2.9	3.1	3.3	3.1	3.3	3.7	3.7	3.3	3.1	(37)	
		MCWBR	6.7	6.9	7.4	7.6	7.5	7.4	7.7	8.1	7.7	7.1	6.6	6.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.398	0.431	0.466	0.511	0.473	0.461	0.454	0.452	0.449	0.413	0.401	0.401	(40)	
		taud	2.497	2.388	2.293	2.157	2.308	2.343	2.372	2.379	2.364	2.497	2.528	2.514	(41)	
	Ebn at Noon		909	888	859	813	829	828	837	850	862	894	899	897	(42)	
		Edh at Noon	107	123	138	157	132	126	123	124	126	109	103	104	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.65	6.05	6.04	5.85	5.39	5.59	5.80	5.76	5.44	4.91	4.79	5.15	(44)	
	RadStd		0.15	0.24	0.28	0.28	0.33	0.30	0.31	0.34	0.29	0.27	0.44	0.34	(45)	

Historical Trends

	DBAvg	Heating
--	-------	---------